

Ch 9 Problems: p.254 # P9-4, P9-6, P9-8, P9-10, P9-14

- **Q9-4.** Is the size of the ARP packet fixed? Explain.

ARP Packet is not fixed because the length of the source/destination MAC and IP addresses are variable.

- **Q9-6.** Assume we have an isolated link (not connected to any other link) such as a private network in a company. Do we still need addresses in both the network layer and the data-link layer? Explain.

Yes, for application layer functionality. If they don't need internet access then no network layer needed but that would mean the layers above the network layer wouldn't work like browsers, OS updates, anything requiring internet access would be broken. would require specialized application software to communicate between the 2 nodes using data link layer addressing only.

- **Q9-8.** In Figure 9.9, why is the destination hardware address of the frame from A to B a broadcast address?

Broadcast used to find all the devices on the network and to build routing table entries for forwarding function used for next hop packet forwarding.

- **Q9-10.** When we talk about the broadcast address in a link, do we mean sending a message to all hosts and routers in the link or to all hosts and routers in the Internet? In other words, does a broadcast address have a local jurisdiction or a universal jurisdiction? Explain.

No, internet is one layer above the data link layer. DLL is broadcasting to the MAC layer addressing network layer. The L2 layer.

- **Q9-14.** How many IP addresses and how many link-layer addresses should a router have when it is connected to five links?

This question is ambiguous. A router is a hardware and sw box. Routers have ports with ethernet network chips which are the equivalent of NIC cards in a PC. Each NIC chip has a MAC address. If there are 5 ports then there are 5 NIC chips or 5 link layer addresses. IP addresses live above the Link layer and are owned by hosts. The router has a single IP address to connect to the internet for network administration and sw updates. It isn't clear if the 5 links includes the router network interface. If so then there are 4 ports, if not then the router has 5 ports.

If there are 4/5 ports then there are 4/5 NIC chips. This does **not** imply there are 4/5 IP addresses in the router as the question asks.

The router holds IP addresses in a routing table. The IP addresses in the routing table are comprised of source and destination addresses. The number of IP addresses in the routing table are dynamic. On power up the routing table is blank. Once it starts it builds a routing table. The router forwards incoming packets by examining the destination address and forwards the packet to the link indicated by the routing table entry. If there are 4/5 links then there is a minimum of 4/5 entries in the routing table.

Link layer addresses are assigned to the hardware at time of shipping. For 5 links there are 5 link layer addresses.

IP addresses are not assigned to the hardware. This is dynamically determined.

5 link layer addresses and a minimum of 1 IP address depending on the state of the IP routing table.